ELEOS Recumbent Off-Road Handcycle

An extremely detailed website content document for explaining the need, design, prototype, testing, and future direction of an e-assist off-road handcycle.

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Website goal: If someone has never heard of a handcycle, ELEOS Solutions, e-assist, double wishbone suspension, or this senior design project, this page should give them enough context to understand the problem, the design, what was built, what worked, what did not, and what should happen next.

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1. Executive Summary

The ELEOS Recumbent Off-Road Handcycle is a senior design engineering project created to address a practical mobility gap: rugged, off-road mobility devices exist, but they are often expensive, specialized, and difficult to manufacture or repair. The project focused on designing and building a functional e-assist handcycle that could operate on difficult terrain while remaining far more accessible than many commercial off-road handcycles.

The prototype is a recumbent, three-wheeled handcycle with two front wheels and one driven rear wheel. The rider sits low in a reclined position and propels the vehicle using hand cranks. Electric assist supplements the rider's input, helping with acceleration, hills, and longer rides. The final prototype integrates a custom frame, front double wishbone suspension, steering linkage, single-sided front wheel mounts, mechanical braking, rear drivetrain, and a commercial e-bike assist kit.

The project was completed for ELEOS Solutions, a nonprofit focused on sustainable, human-centered engineering solutions for underserved populations. The broader purpose was not simply to build a recreational machine, but to explore a lower-cost, repeatable design path for mobility devices that could eventually serve both U.S. recreational users and users in low-resource environments.

In plain language, the project asked: can a small student engineering team build a working off-road handcycle that is durable, rideable, manufacturable, and meaningfully cheaper than high-end commercial alternatives? The answer from Prototype 1 is yes, with important refinements still needed before a final optimized version.

Quick project snapshot

Category	Current Prototype Summary
Vehicle type	Recumbent off-road handcycle with integrated e-assist
Propulsion	Hand cranks plus rear-wheel drivetrain; electric assist integrated into drivetrain
Wheel layout	Two front wheels, one rear drive wheel
Suspension	Double wishbone front suspension with air shocks; rear suspension inherited/adapted from bike components
Manufacturing approach	In-house fabrication using welding, machining, hand tools, standard bike tools, 3D printing, and some cast/machined parts
Testing status	Successfully tested on gravel, uneven terrain, and road conditions; approximately 15 mph off-road and 25-30 mph on road
Current cost status	Approximately $3050 in material/component cost, above the $2500 MVP target but still far below high-end commercial off-road handcycles

2. The Human Need

For many people, mobility is not just about getting from one place to another. It is tied to independence, dignity, recreation, employment, community participation, and the ability to move through the real environments where life happens. Standard wheelchairs work well in many indoor or paved environments, but they can struggle on gravel, dirt, grass, sand, mud, rocks, inclines, and uneven rural paths.

The need becomes even more significant in rural areas and low- and middle-income countries. In many of these environments, paved sidewalks, accessible transportation, and smooth roads are not guaranteed. A mobility device that only works on flat pavement can severely limit a user's independence. A rugged, off-road capable device can expand where a person can go and what they can do.

Commercial off-road handcycles exist, but they can be prohibitively expensive. The project report compared available products and noted that one high-end off-road solution, the Reactive Adaptations Mako, exceeds $17,800, while a lower-cost Amtryke option still did not satisfy all requirements because it lacked off-road capability and suspension. That gap helped define the opportunity for this project: a lower-cost, functional, rugged handcycle that could be built, repaired, and improved without relying entirely on premium proprietary systems.

The project was also motivated by two user stories:

- A disabled person who enjoys the outdoors wants a mountain-bike-equivalent wheelchair so they can participate in a hobby they enjoy.
- A disabled person in a low-resource or rural environment wants an off-road-capable wheelchair so they can participate in the community and provide for themselves.

These stories shaped the design direction. The project was never only about building a trike. It was about building a practical platform that makes rugged mobility more attainable.

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3. What a Handcycle Is

A handcycle is similar to a bicycle, but instead of using pedals operated by the rider's legs, it uses cranks operated by the rider's arms. The rider turns the cranks with their hands, and that rotary motion is transferred through a chain or gear system to drive one or more wheels. Handcycles are commonly used by people who have limited or no use of their legs, but they can also be used for adaptive recreation, fitness, transportation, and racing.

A recumbent handcycle places the rider in a seated or reclined position rather than an upright bicycle posture. This is helpful because it lowers the center of gravity, improves comfort, and provides a stable base for riders who may not have full trunk or lower-body control. The ELEOS design is also inverted relative to a conventional delta trike layout: it uses two wheels at the front for steering and stability, with the rear wheel serving as the drive wheel.

For a website audience, it is useful to describe the system in four simple pieces:

Part of the handcycle	Simple explanation
Rider interface	The seat, leg supports, belt/straps, and handle/crank positions that allow the rider to sit securely and operate the vehicle.
Steering system	The handlebars, steering column, tie rods, kingpins, and wheel mounts that turn the front wheels.
Drivetrain	The hand cranks, sprockets, chains, e-assist motor, cassette, and rear wheel that turn arm motion into forward motion.
Suspension and structure	The frame, control arms, shocks, and mounts that support loads and absorb bumps on rough terrain.

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4. Client and Project Context

The project partner, ELEOS Solutions, is described in the project report as a faith-driven nonprofit dedicated to serving underserved populations through innovative and sustainable engineering solutions. The organization emphasizes servant leadership, collaboration, stewardship, and practical designs that are simple, affordable, repeatable, and manufacturable in low-resource settings.

For this project, ELEOS Solutions challenged the senior design team to develop a recumbent off-road handcycle platform with integrated power assist. The initial vision included an urban/recreational version with gear shifting and e-assist, as well as a simplified low-cost design path that could eventually be adapted for lower-resource settings.

The team consisted of mechanical engineering and electrical engineering students, with Sam Manina serving as project manager. The technical scope included mechanical design, CAD modeling, suspension and steering geometry, drivetrain prototyping, welding and fabrication, e-assist integration, testing, and documentation handoff.

5. Requirements and Success Criteria

The client's minimum viable project criteria were intentionally practical. The project would not be judged only by whether the CAD model looked good, but by whether the prototype could actually be built, tested, and evaluated against real use cases.

MVP criterion	What success meant	How Prototype 1 performed
Off-road capable	Successfully operate the final prototype over difficult terrain.	Met in functional testing. The prototype operated on gravel and uneven terrain, reaching approximately 15 mph off-road.
Accessible cost	Sum the price of final prototype components and compare to a maximum of $2500.	Partially met. The final material cost was approximately $3050, above the original target but still substantially less than many commercial off-road handcycles.
DIY capable	Successfully manufacture a prototype using only hand tools, welding, and bike tools.	Substantially met. Prototype 1 was fabricated using accessible in-house processes, standard bike tools, welding, machining, 3D printing, and available shop resources.

The team also developed additional specifications around hill climbing, battery life, flat-ground testing, descending, and trail or gravel testing. These tests helped turn broad goals into measurable engineering outcomes.

6. System Architecture

The final prototype can be understood as a collection of interconnected subsystems. Each subsystem had its own design constraints, but the hardest problems came from the interactions between them. For example, the front drivetrain had to pass through or around the steering system; the steering system had to move with the suspension; the suspension loads affected the wheel mounts; and the brake mounts had to fit around the kingpin, wheel, and tie rods.

Subsystem	Primary function	Key engineering concern
Frame	Supports rider, drivetrain, steering, suspension, seat, and e-assist components.	Strength, weldability, geometry, weight, and manufacturability.
Rider interface	Positions and secures rider while allowing arm-powered propulsion.	Comfort, reach, adjustability, safety, and transfer accessibility.
Steering	Turns the two front wheels through a steering column and tie rods.	Geometry, caster, camber, scrub radius, bump steer, and steering sensitivity.
Suspension	Absorbs off-road impacts while maintaining wheel contact.	Shock stiffness, geometry, wheel travel, load paths, and mount strength.

Wheel mounts	Support single-sided front wheels and interface with kingpins and brakes.	High loads, rigidity, brake reaction forces, looseness, and manufacturability.
Drivetrain	Transfers human input from hand cranks to the rear wheel.	Chain alignment, derailment prevention, sprocket layout, tensioning, and steering interaction.
E-assist	Adds motor power and electronic control to reduce rider effort.	Motor mounting, wiring, sensors, battery, controls, and safe integration.

7. Frame and Rider Interface

The frame is the base structure for the entire vehicle. It sets the rider position, wheelbase, steering column location, suspension pickup points, rear drivetrain location, and e-assist mounting. Because so many subsystem dimensions depend on the frame, mistakes in the frame geometry can force redesign across the whole vehicle.

Prototype 1 used square tubing because it was easier to weld, fixture, and align within the available fabrication environment. Earlier design thinking considered round tubing, which is common in many bicycle and vehicle frames, but square tubing reduced manufacturing complexity for the first full prototype. This was a practical tradeoff: square tubing helped the team build a working vehicle more quickly, while future teams can pursue round tubing to reduce weight and improve aesthetics after the geometry is proven.

The rider interface included a bucket-style seat, leg supports/slings, straps, and handlebars. Early assembly was used to confirm that a person could sit in the vehicle and comfortably reach the cranks. This mattered because ergonomic fit is not something that should be left until the end of a mobility project. If the user cannot comfortably operate the system, the mechanical design is not successful even if it is structurally strong.

Major frame and rider-interface design considerations:

- The frame needed to be strong enough for off-road loads while remaining light enough to move and transport.
- The seat needed to secure the rider laterally and prevent sliding during turning or rough terrain.
- Leg supports needed to hold the rider's legs in place and prevent bouncing out over bumps.
- The crank and handlebar reach needed to allow comfortable arm motion and steering control.
- The design needed to remain manufacturable with accessible tools rather than relying on expensive custom production methods.

Prototype 1 is intentionally overbuilt. That made sense for safety during initial testing, but it also creates a clear optimization path for Prototype 2: calculate actual loads, perform more meaningful FEA, and reduce weight while preserving strength.

8. Steering System

The steering system controls the two front wheels. Because the vehicle has two front wheels rather than one bicycle fork, steering is more complex than a conventional bike. Both front wheels must turn together, maintain reasonable alignment, and continue to behave well as the suspension moves.

The prototype uses a steering column connected to a central pitman arm. The pitman arm pushes and pulls tie rods, which transmit motion to kingpin pitman arms at the wheel assemblies. The kingpins rotate inside their mounts, causing the front wheels to turn. This creates a mechanically direct steering system, similar in concept to go-kart or recumbent trike steering.

Important steering terms explained for non-engineers:

Term	Plain-language meaning	Why it matters
Kingpin	The vertical-ish pivot that lets the front wheel turn left and right.	It defines the steering axis and strongly affects feel, stability, and manufacturability.
Caster	How much the steering axis tilts forward or backward from the side view.	More caster tends to make the wheels self-center, improving stability but increasing steering effort. Prototype 1 used roughly 9 degrees.
Camber	How much the wheel leans inward or outward from the front view.	It affects tire contact, handling, and rollover behavior. Prototype 1 was close to neutral.
Scrub radius	The distance between where the tire contacts the ground and where the steering axis hits the ground.	Too much scrub radius makes steering harder and increases tire wear.
KPI	Kingpin inclination angle, or how much the kingpin leans inward from the front view.	A major way to tune scrub radius without changing the wheel size.
Bump steer	Unwanted steering caused by suspension movement.	A safety issue that can be reduced by keeping tie rod motion aligned with control arm motion.

The steering handoff guide highlights that there are many angles and geometries to consider and recommends that future redesigns account for caster, camber, scrub radius, KPI, and bump steer. Prototype 1 showed that the steering worked and produced a tight turning radius, but it may be too sensitive at higher speeds. Future teams should consider shortening the pitman arms or otherwise reducing steering sensitivity.

9. Front Suspension and Shocks

The front suspension is one of the most important systems for off-road capability. Its job is to absorb impacts, maintain tire contact with uneven ground, improve stability, and reduce the loads transmitted directly into the rider and frame. Without suspension, an off-road handcycle can become uncomfortable, hard to control, or structurally overstressed.

Prototype 1 uses a double wishbone style front suspension. In this layout, an upper and lower control arm connect the wheel assembly to the frame. A shock absorber mounts between the suspension linkage and the frame/control arm system. As the wheel moves upward over a bump, the control arms guide the wheel path while the shock resists and damps the motion.

The team selected this architecture because it provided a strong balance between off-road performance, tunability, and manufacturability. It also allowed the team to design around off-the-shelf control arms and commercially available shocks, rather than fabricating every part from scratch.

Shock selection was a critical design decision. If the shock is too stiff, the wheel does not travel enough and the suspension fails to absorb terrain. If the shock is too soft, the system can bottom out, which means the suspension reaches the end of its travel and transfers impact loads directly into the frame. The team modeled the suspension at equilibrium to estimate forces and select shock parameters rather than guessing.

Prototype 1 proved the suspension architecture could be integrated into the vehicle, but testing and post-project review suggest the front suspension is too stiff and does not travel enough. This is one of the most important areas for future optimization.

10. Wheel, Kingpin, and Brake Mounting

The front wheels are single-sided supported. Unlike a traditional bicycle front wheel, which is supported on both sides by a fork, each front wheel on this vehicle mounts from only one side. This is required by the double wishbone/kingpin layout, but it increases the importance of the wheel mounts, axle interface, kingpin fit, and brake mount stiffness.

Single-sided support creates higher bending loads and makes looseness more noticeable. If a front wheel mount is not stiff enough, the wheel can wobble, change alignment, interfere with other components, or reduce rider confidence. If a brake mount is not stiff or strong enough, braking forces can deform or break the bracket.

The team learned this directly during prototyping. Initial brake caliper mountings made from aluminum failed because of the force required to stop the front wheels. The team redesigned the mounts and changed to a higher-strength, stress-resistant steel configuration. The revised mounts have performed significantly better in testing.

Future work should still validate the brake and wheel mounts more rigorously. The Prototype 2 issues list specifically recommends remaking loose kingpin mounts so they fit flush and can be tightened down, changing both brake mounts so they sit to the rear of the kingpin to avoid tie rod interference, and verifying that the brake mount design is actually strong enough for repeated use.

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11. Drivetrain and Chain Routing

The drivetrain transfers power from the rider's arms to the rear wheel. The system begins at the hand cranks near the rider's upper body. From there, the chain travels down the steering column region and back along the frame. It connects through the e-assist module, then transfers power to the rear cassette and rear wheel.

This system was one of the most challenging parts of the design because the hand cranks and steering system occupy the same physical area. The rider needs to crank and steer, and the chain must continue to run without derailing as the steering angle changes. The gear train handoff document explains that sprocket placement was driven by interference prevention, chain alignment, and steering interaction.

The final drivetrain uses standardized bicycle components wherever possible: crank gear, turn sprockets, rear cassette, bike chain, tensioners, and master links. Standard parts reduced cost, simplified replacement, and made the system more understandable for future teams.

Key drivetrain lessons:

- Chain alignment is the most important factor in preventing derailment.
- Looking down the chain path, the chain should be as straight as possible through most of its motion.
- Straight-line chain path checks in CAD should include steering extremes, not only the straight-ahead position.
- Stacking the two turn sprockets on top of each other caused misalignment and was not workable.
- Chain guards are important for safety, especially near the rider's legs and hand cranks.
- A simpler turn sprocket interface would reduce manufacturing time and future maintenance difficulty.

Prototype 1 proved the drivetrain concept, but it still has refinement needs. The front drivetrain can derail during left turns because it becomes loose when sized for full right steering. Future work should add better tensioning or redesign the turn sprocket/tensioner geometry.

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12. E-Assist System

The e-assist system helps reduce rider effort, especially during starts, hill climbs, and longer rides. For users with limited upper-body strength or endurance, e-assist can make the difference between a device that is technically rideable and a device that is practically useful.

The prototype uses a standardized commercial e-bike assist kit rather than a fully custom motor system. This was intentional. A commercial kit reduces electrical design complexity, improves part availability, and gives the project a more realistic path toward repeatability. The integrated system includes a motor, battery, LCD screen, throttle, pedal assist system, torque sensor, shift sensor, brake sensor, and speed sensor.

The e-assist system is integrated into the drivetrain so rider input and motor power work together to drive the rear wheel. The system is designed to support pedal assist operation as well as throttle-driven assistance. In final testing, the e-assist was fully integrated with working speed, gear shift, and braking sensors.

One recommendation for future teams is to consider a less powerful motor. A 750 W motor provides strong assist, but it may be more power than needed for the final use case. A smaller motor could reduce weight, cost, packaging difficulty, and safety concerns while still providing enough help for climbing and rough terrain.

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13. Manufacturing and Prototype Build

The prototype was built using an intentionally accessible manufacturing strategy. The frame was welded, custom components were machined, some parts were produced through 3D printing or casting-related methods, and standard bicycle components were integrated wherever possible. This approach aligned with the DIY-capable requirement: the project should be buildable using tools and processes that are more accessible than high-end manufacturing equipment.

The build process followed an engineering progression: research, CAD concept, subsystem prototyping, component ordering, frame fabrication, subsystem assembly, fit validation, full integration, testing, and revision. This progression was not perfectly linear. Many parts had to be adjusted after testing exposed issues that were not obvious in CAD.

Prototype 1 included an early assembly phase before the full drivetrain was installed. This was used to check rider fit, seat position, crank reach, and basic ergonomics. Because the team had planned and sized the system carefully, the initial fit was good and did not require major redesign. That was a major success: it showed that upfront layout planning saved time during physical assembly.

Manufacturing lessons for future teams:

- Design for manufacturability should happen before parts are fully committed to CAD and fabrication.
- Mounting tabs, brackets, and wheel interfaces should be designed around how they will actually be cut, drilled, welded, and assembled.
- Prototype parts should be simple enough to remake quickly after testing reveals problems.
- Material changes should be driven by load paths, not only availability. The brake mount redesign from aluminum to steel is a good example.
- The next frame should likely use round tubing after more simulation, but square tubing was the right practical choice for getting Prototype 1 built.

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14. Testing and Validation

Testing transformed the project from a collection of CAD models and assemblies into a real mobility device. The team tested function, fit, steering, braking, drivetrain behavior, e-assist integration, and off-road performance.

The project report outlined several planned test categories: control tests, realistic flat-ground testing, elevation gain, elevation descent, and trail or gravel testing. These tests were intended to verify that the handcycle could function as a complete system rather than only as isolated subsystems.

Final prototype testing showed that the handcycle operated successfully on real off-road terrain, including gravel and uneven surfaces. The vehicle reached approximately 15 mph on gravel and approximately 25-30 mph on pavement. These results validated the basic off-road capability of the platform and demonstrated that the e-assist, drivetrain, steering, and frame could function together.

Testing also revealed important limitations. The front suspension is too stiff, some drivetrain derailment can occur, wheel and kingpin mount looseness should be addressed, and braking/mounting components should receive more validation. These are not failures of the project; they are the exact purpose of a first prototype. Prototype 1 proved the concept and identified what must be improved next.

15. Cost and Accessibility

The original cost target was $2500 for final prototype components. Prototype 1 ended at approximately $3050 in material/component cost. That means the prototype exceeded the strict MVP cost target, but it still represents a major reduction compared with high-end commercial off-road handcycles that can exceed $17,800.

The current cost should be interpreted carefully. A first prototype often costs more than a refined design because it includes one-off purchases, extra parts, design changes, low-volume sourcing, and components selected for speed and availability rather than production efficiency. Future optimization could reduce cost by simplifying brackets, standardizing brake components, selecting a smaller motor, reusing fewer custom parts, and improving supply planning.

Cost category	Approximate cost in final presentation
Frame	$351.67
Drive train assembly	$1134.22
Steering assembly	$206.81
Wheel mounts	$736.38
Suspension assembly	$264.21
Other components	$360
Total estimated material cost	~$3050, not including manufacturing costs

The biggest cost opportunities appear to be drivetrain simplification, wheel mount refinement, motor/e-assist sizing, and reducing custom-machined or overbuilt components.

16. What Worked and What Needs Refinement

Prototype 1 was successful because it became a real, rideable, off-road-tested machine. It integrated major subsystems that had previously existed separately and showed that the overall architecture is viable. At the same time, it revealed the issues a future team should address before the design becomes a more polished final product.

Area	What worked	What needs refinement
Frame	Strong, buildable, and effective as a first full prototype structure.	Overbuilt and likely heavier than necessary; round tubing and more FEA should be considered.

Steering	Tight turning radius and functional linkage system.	May be too sensitive at high speed; pitman arm length and geometry should be reviewed.
Suspension	Double wishbone architecture was integrated into the vehicle.	Front suspension is too stiff and lacks travel; rear suspension may also be too stiff.
Drivetrain	Rear-wheel drive with e-assist integration worked and used standard bike components.	Front chain can derail during steering; tensioning and chain guards need improvement.
E-assist	Motor, sensors, throttle/assist functions, and drivetrain integration worked.	Motor may be more powerful than needed; packaging and control simplification could help.
Brakes and wheel mounts	Redesigned steel brake mounts worked better than initial aluminum mounts.	Mount validation, kingpin fit, and caliper placement need improvement.
Rider interface	Initial fit and crank reach worked well because of upfront planning.	Seat, transfer access, handholds, and comfort should be improved.

17. Future Work for Prototype 2

The next senior design team should treat Prototype 1 as a proven platform, not a finished product. The hard part of making the overall architecture work has been done. The next phase should focus on optimization, refinement, safety, comfort, and repeatability.

Recommended priorities for Prototype 2:

- Rework suspension tuning. The front suspension should be redesigned or retuned to allow more effective travel. The rear suspension should also be evaluated and potentially softened or replaced.
- Optimize the frame. The current frame is overbuilt. Future teams should calculate loads, run meaningful FEA, and transition toward round tubing if feasible.
- Improve drivetrain tensioning. The front drivetrain needs better slack management through the full steering range, especially during left turns.
- Add chain guards. Guards around the hand cranks, turn sprockets, and leg area are important for safety and clothing protection.
- Refine wheel and kingpin mounts. The kingpins should fit flush and tighten securely; looseness should be removed.
- Standardize brake mounting. Use matching calipers if possible and place both brake mounts behind the kingpin to reduce tie rod interference.
- Consider hydraulic brakes. Hydraulic brakes could improve stopping performance, feel, and reliability for off-road use.
- Reduce steering sensitivity. Shorter pitman arms or adjusted steering geometry may make the vehicle safer at higher speeds.
- Improve rider access and comfort. Add better handholds for transfers, refine the seat, adjust handlebars, and improve strap/leg support systems.
- Add safety accessories. Rearview mirrors, bell or horn, reflectors, rear running lights, parking brake, padding, and possibly a harness should be evaluated.

18. How the Prototype Works - Step by Step

This section explains the vehicle as if the reader is standing next to it and watching someone ride it for the first time. The rider transfers into the recumbent seat, secures their legs in the leg supports, and places their hands on either the crank handles or the auxiliary handlebars. The rider can propel the vehicle by turning the hand cranks, steer with the hand-crank assembly or the auxiliary handlebars, and use e-assist to reduce effort.

Step 1 - Rider setup. The rider sits in the bucket-style seat and positions their legs in the front supports. The low seating position helps stability and keeps the center of gravity lower than a tall upright design. Straps and future harness improvements are important because off-road bumps can lift or shift the rider if they are not secured.

Step 2 - Human power input. The rider turns the hand cranks. This is the equivalent of pedaling a bicycle, except the motion comes from the arms. The crank gear drives a chain downward through the front drivetrain routing system.

Step 3 - Steering interaction. Because the hand cranks are part of the front area of the vehicle, the drivetrain must coexist with steering. The design uses turn sprockets and carefully aligned chain routing so the chain can continue transmitting power while the steering angle changes.

Step 4 - E-assist power addition. The chain path interfaces with the e-assist system. The motor can add power based on throttle or pedal-assist style input, reducing the load on the rider. Sensors such as brake and shift sensors help the system behave more safely and smoothly.

Step 5 - Rear wheel drive. Power travels through the rear drivetrain to the rear cassette and rear wheel. This rear-wheel-drive layout helps place propulsion at the back while allowing the two front wheels to focus on steering, suspension, and braking.

Step 6 - Suspension and steering response. As the front wheels hit uneven terrain, the double wishbone suspension allows controlled vertical movement. The steering linkage and tie rods must continue to work through that movement without creating unsafe bump steer.

Step 7 - Braking and control. The brake system slows the front wheels. Brake mounts are highly loaded because stopping forces can be large, especially when the rider and vehicle are moving downhill or on gravel. This is why the brake mount redesign from aluminum to steel was a meaningful prototype improvement.

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19. Project Development Timeline

A website page benefits from showing that the prototype was not built all at once. The project moved through research, concept development, subsystem design, prototyping, fabrication, testing, and handoff. The timeline below gives a reader a clear sense of how the team progressed from a broad need to a working vehicle.

Phase	Approximate timing	What happened	Why it mattered
Project definition	Early fall	The team met with the client, clarified goals, and defined the problem around low-cost off-road mobility.	This established that success meant more than a functional prototype; it also needed to be accessible and manufacturable.
Research and benchmarking	September	The team researched commercial handcycles, major subsystems, standards, costs, and off-road mobility products.	This showed that available off-road handcycles were expensive and that existing low-cost options did not satisfy the off-road/suspension requirement.
Initial CAD and subsystem concepts	Late September to October	The drivetrain concept, front suspension geometry, steering assembly, and frame ideas were modeled and reviewed.	This helped locate subsystem conflicts early, especially where steering, chain routing, and front suspension overlapped.

Subsystem prototyping	October to November	The drivetrain/steering prototype was built and iterated, shock options were evaluated, and off-the-shelf components were selected.	Physical prototypes revealed practical problems that were not obvious in CAD, especially chain alignment and steering interaction.
Frame and component design	November to winter	The team moved toward a full frame design, refined wheel mounts, integrated a rear bike frame, and prepared purchasing and fabrication documents.	This turned conceptual subsystems into parts that could actually be manufactured and assembled.
Build season	Spring	The frame was fabricated, components were assembled, e-assist was integrated, and brake/wheel mount issues were revised.	This was the shift from design work to real mechanical integration.
Testing and validation	Late spring	The handcycle was ridden on gravel, uneven terrain, and road surfaces. Speeds of roughly 15 mph off-road and 25-30 mph on road were observed.	Testing demonstrated the concept worked and also identified the highest-priority improvements for Prototype 2.
Handoff	End of project	The team created drivetrain, steering/suspension, and issue-tracking documents for the next senior design team.	This allows future work to start from an informed platform instead of rediscovering the same problems.

20. Detailed Component and Subsystem Inventory

This inventory is written for a website visitor who wants to understand what physical parts make up the vehicle. It is not intended to be a full bill of materials, but it explains the role of each major component group.

Component group	Parts included	Function
Main frame	Welded square tube members, cross members, mounting plates, suspension pickup points, rear structure integration.	Carries the rider, connects all subsystems, and defines the core geometry of the handcycle.
Seat and rider support	Bucket seat, seat belt, leg slings/supports, straps, potential future handholds.	Positions and secures the rider so they can crank, steer, and remain stable over bumps.
Front suspension	Upper control arms, lower control arms, suspension tabs, shocks, wheel/kingpin mounts.	Allows the front wheels to move over uneven terrain while maintaining steering and traction.
Wheel/kingpin assemblies	Single-sided front wheels, axles, kingpins, kingpin mounts, brake mounts.	Support the wheels, allow steering rotation, and transmit braking and terrain loads into the suspension.
Steering system	Steering column, handlebars, central pitman arm, tie rods, kingpin pitman arms.	Turns both front wheels together and gives the rider directional control.
Front drivetrain	Hand cranks, crank gear, chain, turn sprockets, guides/tensioning components.	Turns arm motion into chain motion while still allowing steering.
E-assist module	Motor, controller, battery, display, throttle, speed sensor, brake sensors, shift sensor.	Adds electric power and control features to reduce rider effort and improve climbing ability.
Rear drivetrain	Rear cassette, derailleur, chain, rear wheel, rear frame/fork section.	Transfers combined rider and motor power to the rear wheel.

Braking system	Brake levers, calipers, rotors, mounts, cables or future hydraulic lines.	Slows or stops the vehicle; future hydraulic brakes are recommended.

21. Standards, Safety, and Ethical Considerations

Because this is a mobility device, safety is not an optional add-on. The project referenced bicycle and e-bike related standards and considered mountain bike performance requirements, headset standards, consumer product requirements, Oregon bike laws, and electrically power-assisted cycle guidance. These references helped frame the design, even though the prototype is not a commercial certified product.

The safety challenge is that the handcycle combines features from several categories: bicycle, adaptive mobility device, off-road vehicle, and e-assist cycle. That means the team had to think about both mechanical safety and user safety. Mechanical safety includes frame strength, brake strength, steering stability, suspension load paths, chain containment, and wheel mounting. User safety includes rider retention, transfer accessibility, body positioning, visibility, and protection from moving drivetrain components.

Important safety improvements for future development include chain guards, better brake validation, a parking brake, rearview mirrors, a horn or bell, reflectors, lights, improved seat belt or harness strategy, safer top sprocket/hand crank protection, and more refined transfer handholds. These improvements are especially important if the design moves beyond a student prototype into real user testing.

Ethically, the project must be honest about what Prototype 1 is and is not. It is a functional proof-of-concept and working prototype. It is not yet a fully optimized commercial medical or consumer mobility product. The responsible next step is to continue testing, document limitations clearly, and improve safety-critical areas before broader deployment.

22. Full Website Narrative Draft

The following is longer-form website copy that can be adapted directly into a portfolio page, project page, or team website. It is written in a public-facing tone and assumes the reader may not be familiar with senior design, handcycles, or mobility engineering.

Opening paragraph: The ELEOS Recumbent Off-Road Handcycle is a student-engineered mobility platform designed to make rough-terrain travel more accessible. Built as a senior design project at George Fox University, the prototype combines arm-powered propulsion, electric assist, custom steering, front suspension, and a manufacturable welded frame into one working vehicle. The goal was simple but challenging: build a handcycle that could handle gravel and uneven terrain while staying far more affordable and more buildable than many commercial off-road handcycles.

Problem paragraph: For many wheelchair users, rough terrain creates a boundary. Dirt paths, gravel roads, hills, grass, and uneven rural surfaces can limit access to outdoor recreation, work, school, and community life. Commercial off-road handcycles can solve part of this problem, but many are expensive and difficult to manufacture or repair. This project explored whether an engineering team could create a lower-cost, off-road-capable alternative using standard bicycle parts, accessible fabrication methods, and a modular design that future teams could continue improving.

Solution paragraph: The final prototype is a recumbent, three-wheeled handcycle with integrated electric assist. The rider sits low in the frame, turns hand cranks to drive the rear wheel, and uses the e-assist system for additional power. Two front wheels provide steering and suspension support, while the rear wheel delivers propulsion through a bike-style drivetrain. The vehicle was tested on gravel and

pavement and reached approximately 15 mph off-road and 25-30 mph on road.

Engineering paragraph: The project required the team to solve several interconnected mechanical problems. The steering system had to turn two front wheels while moving with suspension travel. The drivetrain had to route chain motion from the hand cranks through the steering area and back to the rear wheel. The suspension had to absorb off-road impacts without bottoming out or becoming too stiff. The wheel and brake mounts had to handle single-sided wheel support and high braking loads. Each subsystem had to work not only by itself, but as part of one complete machine.

Outcome paragraph: Prototype 1 successfully demonstrated the core concept. It was rideable, off-road capable, and fully integrated with working drivetrain, steering, suspension, and e-assist systems. It also revealed clear next steps: reduce frame weight, improve suspension travel, refine wheel and brake mounts, add chain guards, improve rider comfort, and simplify the drivetrain for manufacturing. The project now provides a strong foundation for the next senior design team to optimize rather than start over.

23. Recommended Images and Captions for the Website

The website page should be visual. The best images are not generic stock photos; they are real project visuals that show the progression from problem to prototype. Suggested images and captions are listed below.

Image type	Where to use it	Suggested caption
Final prototype hero photo	Top of page	Final ELEOS off-road handcycle prototype, built and tested by the 2025-26 senior design team.
Team photo	Client/team section	The ELEOS Handcycle senior design team at George Fox University.
Commercial handcycle price comparison	Problem section	Existing off-road handcycles can cost many thousands of dollars, creating a major accessibility barrier.
CAD system overview	Design overview	Subsystem layout showing frame, steering, suspension, wheel mounts, drivetrain, and e-assist.
Suspension close-up	Suspension section	Double wishbone suspension with upper/lower control arms and central shock.
Drivetrain prototype photo	Drivetrain section	Early drivetrain and steering-chain routing prototype used to validate chain motion.
Brake/wheel mount before-after	Iteration section	Brake and wheel mount design was revised after testing revealed strength and rigidity issues.
Testing photo or video still	Testing section	Prototype testing on gravel and uneven terrain validated off-road capability.
Future work graphic	Next steps section	Future teams can optimize weight, suspension travel, brake reliability, drivetrain simplicity, and rider comfort.

24. Website Copy Blocks and Suggested Page Layout

The following is a suggested website page structure. It is written for a general audience, not just engineering faculty. It explains the need first, then gradually introduces the technical details after the reader understands why the project matters.

Website section	Purpose	Suggested content
Hero	Create immediate understanding and interest.	Large photo of final prototype, title, one-sentence summary, and three badges: Off-road capable, E-assist, Built for accessibility.
The problem	Explain why the project matters.	Brief explanation of mobility limitations, high commercial cost, and rough-terrain barriers.
What is a handcycle?	Educate readers who have never seen one.	Simple comparison to bicycle and wheelchair; explain arm-powered propulsion.
Project goals	Show success criteria.	Three cards: Off-road capable, Accessible cost, DIY manufacturable.
Final prototype overview	Show what was built.	Annotated image with frame, steering, suspension, drivetrain, e-assist, and wheel mounts.
Subsystem deep dives	Explain engineering work.	Expandable sections for frame, steering, suspension, drivetrain, e-assist, brakes, and manufacturing.
Testing results	Prove the design worked.	Gravel trail testing, road speed, off-road performance, and what was learned.
What comes next	Position the project as continuing work.	Prototype 2 recommendations and optimization roadmap.
Team and acknowledgments	Credit contributors.	Team members, advisors, client, school, and project role descriptions.

Suggested hero copy

ELEOS Recumbent Off-Road Handcycle
A functional e-assist handcycle prototype built to make off-road mobility more accessible, manufacturable, and affordable.

Suggested section intro

For many wheelchair users, rough terrain creates a hard boundary. Dirt paths, gravel roads, grass, sand, and hills can turn basic movement into a major obstacle. The ELEOS Handcycle project set out to challenge that boundary by creating a rugged, arm-powered, electric-assist handcycle that could be built with accessible manufacturing methods and improved by future teams.

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25. Source Notes

This document was prepared from the project documents provided by the ELEOS Handcycle senior design team. The source material includes:

- Senior Design Project Report - Recumbent Off-Road Handcycle with Integrated E-Assist, 2025-26.
- Steering & Front Suspension Guide.
- Gear Train Design Considerations handoff document.
- Issues to Fix for Prototype 2 handoff document.
- Spring final presentation draft and project discussion notes supplied by the team.

Some values, such as final test speeds and final presentation cost totals, reflect project status as communicated in the final presentation and team notes. They should be updated on the website if the team completes additional testing, weighs the final prototype, changes the bill of materials, or completes Prototype 2.